

Rectangular Schur Polynomials and Cigler’s Conjecture 8

A Laurent–Gram proof of the even-shift Hankel identity

AI-assisted research draft

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Abstract

Let $b_m(t) = \sum_{r \geq 0} \binom{\lfloor m/2 \rfloor}{r} \binom{\lceil m/2 \rceil}{r} t^r$. We prove, for all integers $k, n \geq 0$, the polynomial identity

$$\det(b_{2k+i+j}(t))_{0 \leq i, j < n} = t^{\lfloor n^2/4 \rfloor} s_{(n^k)}(\underbrace{1, \dots, 1}_k, \underbrace{t, \dots, t}_k).$$

The proof realizes the Hankel matrix as a Laurent-polynomial Gram matrix, changes to a consecutive monomial basis, and identifies the resulting Toeplitz determinant by dual Jacobi–Trudi. A positive sum over partitions inside a rectangle proves full coefficient support and palindromicity; restriction of a Schur module to SL_2 proves unimodality. A confluent alternant supplies explicit numerator polynomials of exactly the degrees and with exactly the reciprocity required in Cigler’s Conjecture 8 (arXiv:2111.14492v3). Thus every assertion of that conjecture is obtained, not only its positivity assertion. Further consequences include stable edge coefficients, a dimension product, an exact generic recurrence, and fixed-parameter asymptotics. All identities and boundary cases are proved symbolically; a standard-library Python verifier supplies additional exact finite checks.

Status. This is a complete mathematical proof draft relative to the classical Schur identities and Schur-module character theorem stated below. It has not been independently refereed or checked in a proof assistant. A targeted public literature search did not locate an earlier resolution of the selected conjecture; this is not an exhaustive priority certificate.

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1 The selected problem and its provenance

1.1 Why this is related to, but not a duplicate of, the manifest

The supplied manifest [4] contains an entry entitled *Product formulas for ballot-polynomial Hankel determinants*. Its stated scope is Conjectures 13–15 in Section 5 of Cigler’s paper [1]. The present article instead treats *Conjecture 8 in Section 4*, concerning the polynomial extension $b_m(t)$ of the middle binomial coefficients. The determinant family, normalization, and numbered conjecture are all specified below. None of the manifest’s reports is used as a theorem in the proof. In particular, the manifest’s descriptions of claimed results are not silently treated as independently verified mathematics.

The original source is the third version of arXiv:2111.14492, dated 30 December 2021. The arXiv record consulted on 20 September 2026 still identified that version as current. Conjecture 8 appears on printed pages 16–17, with its numerator formula and reciprocity numbered (56) and (57). A targeted search for the paper title and for combinations of “Cigler”, “Conjecture 8”, “Schur”, and “proof” found no prior resolution. An absence of search results cannot establish novelty. The package’s `sources.md` records the search and its limits.

1.2 Notation and the precise target

For integers $m \geq 0$, define

$$b_m(t) = \sum_{r=0}^{\lfloor m/2 \rfloor} \binom{\lfloor m/2 \rfloor}{r} \binom{\lceil m/2 \rceil}{r} t^r. \quad (1.1)$$

Thus

$$b_0 = b_1 = 1, \quad b_2 = 1 + t, \quad b_3 = 1 + 2t, \quad b_4 = 1 + 4t + t^2.$$

At $t = 1$, Vandermonde’s convolution gives $b_m(1) = \binom{m}{\lfloor m/2 \rfloor}$, explaining the name. For a shift $s \geq 0$ and an order $n \geq 0$, put

$$D_s(n, t) = \det(b_{s+i+j}(t))_{0 \leq i, j < n}, \quad D_s(0, t) = 1. \quad (1.2)$$

The empty determinant convention is important throughout. We will prove

$$D_0(n, t) = t^{\lfloor n^2/4 \rfloor}. \quad (1.3)$$

For even shifts we write

$$R_{k,n}(t) = \frac{D_{2k}(n, t)}{D_0(n, t)} = \frac{D_{2k}(n, t)}{t^{\lfloor n^2/4 \rfloor}}. \quad (1.4)$$

At this point the quotient is interpreted in $\mathbb{Q}(t)$; its polynomiality is a conclusion, not an assumption. In Cigler’s notation, $R_{k,n}(t) = d_{2k}(n, t) = r_{2k}(n, t)$.

For $k \geq 1$, Conjecture 8 asks that $R_{k,n}$ have positive coefficients at every degree from 0 through kn , be palindromic and unimodal, and admit the expression

$$R_{k,n}(t) = \frac{\sum_{j=0}^k (-1)^j c_{2k,j}(n, t) t^{j(n+j)}}{(1-t)^{k^2}}, \quad (1.5)$$

where

$$\deg_t c_{2k,j}(n, t) = 2j(k-j), \quad (1.6)$$

$$c_{2k,k-j}(n, t) = t^{2j(k-j)} c_{2k,j}(n, t^{-1}). \quad (1.7)$$

Here *unimodal* means weakly increasing up to the center and weakly decreasing afterwards; strict unimodality is neither required nor true in general. We prove all of these assertions, including the stated degrees.

2 Schur notation and the classical ingredients

A partition is a weakly decreasing finite list of nonnegative integers, with trailing zeros ignored. Its size is $|\lambda| = \sum_i \lambda_i$, and its conjugate is denoted λ' . The notation (n^k) means the rectangle with k rows, each of length n . If $k = 0$ or $n = 0$, this is the empty partition.

For a list $X = (x_1, \dots, x_m)$, let $e_r(X)$ and $h_r(X)$ be the elementary and complete homogeneous symmetric polynomials:

$$\sum_{r \geq 0} e_r(X) u^r = \prod_{a=1}^m (1 + x_a u), \quad \sum_{r \geq 0} h_r(X) u^r = \prod_{a=1}^m (1 - x_a u)^{-1}. \quad (2.1)$$

Both are zero for negative subscripts and equal one at subscript zero; $e_r(X) = 0$ also when $r > m$.

A semistandard Young tableau has weakly increasing rows and strictly increasing columns. The Schur polynomial $s_\lambda(X)$ is the sum of the weight monomials of all such tableaux of shape λ , with entries from $\{1, \dots, m\}$. The following classical descriptions of the same polynomial will be used [2, 3]:

$$s_\lambda(X) = \det(h_{\lambda_i - i + j}(X))_{1 \leq i, j \leq \ell(\lambda)}, \quad (2.2)$$

$$s_\lambda(X) = \det(e_{\lambda'_i - i + j}(X))_{1 \leq i, j \leq \lambda_1}, \quad (2.3)$$

$$s_\lambda(X) = \frac{\det(x_i^{\lambda_{m+1-a} + a - 1})_{1 \leq i, a \leq m}}{\det(x_i^{a-1})_{1 \leq i, a \leq m}}. \quad (2.4)$$

The last formula is first read for distinct variables and then extended as a polynomial identity. The increasing order of the exponents in (2.4) fixes the signs used later.

We also use one standard representation-theoretic fact. If V is a complex vector space of dimension m , the Schur module $S_\lambda(V)$ is a homogeneous polynomial representation of $\mathrm{GL}(V)$ of degree $|\lambda|$, and the trace of $\mathrm{diag}(x_1, \dots, x_m)$ on it is $s_\lambda(X)$. The tableau basis has exactly the corresponding weight monomials. This character theorem is explicitly stated in Fulton's treatment [2]. It is an imported classical result; it is not claimed as a new construction here. The needed elementary SL_2 argument is proved in Section 5.

For our repeated-variable specialization, write

$$X_k = (\underbrace{1, \dots, 1}_k, \underbrace{t, \dots, t}_k). \quad (2.5)$$

For example,

$$e_r(X_k) = \sum_a \binom{k}{a} \binom{k}{r-a} t^a, \quad (2.6)$$

$$h_r(X_k) = \sum_{a=0}^r \binom{k+a-1}{a} \binom{k+r-a-1}{r-a} t^a \quad (k \geq 1, r \geq 0). \quad (2.7)$$

We use the usual convention that a binomial coefficient with an invalid lower index is zero.

3 The central determinant identity

Theorem 3.1 (Even-shift Hankel–Schur identity). *For all integers $k, n \geq 0$,*

$$\boxed{D_{2k}(n, t) = t^{\lfloor n^2/4 \rfloor} s_{(n^k)}(X_k)}. \quad (3.1)$$

The identity holds in $\mathbb{Z}[t]$. In particular, $D_0(n, t) = t^{\lfloor n^2/4 \rfloor}$ and $R_{k,n}(t) = s_{(n^k)}(X_k)$.

The determinant on the left has entries depending on $i + j$. The proof turns it into a determinant depending on $i - j$. The entire normalizing power of t comes from the change of basis, which is why both parities of the matrix order must be tracked explicitly.

3.1 A constant-term model for the moments

Work temporarily over $K = \mathbb{Q}(t)$ and in the Laurent-polynomial ring $K[z, z^{-1}]$. Define

$$A(z) = (1+z)(1+t/z) = 1+t+z+t/z. \quad (3.2)$$

Let CT_z extract the coefficient of z^0 . Substitution $z \mapsto t/z$ defines a K -linear ring involution $f \mapsto f^*$. It fixes A , and

$$\text{CT}_z f^* = \text{CT}_z f. \quad (3.3)$$

Indeed, the coefficient of z^0 is unchanged when each monomial z^r is replaced by $t^r z^{-r}$.

Lemma 3.2. *For $a \geq 0$,*

$$b_{2a}(t) = \text{CT}_z A^a, \quad b_{2a+1}(t) = \text{CT}_z(1+z)A^a = \text{CT}_z(1+t/z)A^a. \quad (3.4)$$

Proof. Expand $A^a = (1+z)^a(1+t/z)^a$. Its constant term is $\sum_r \binom{a}{r}^2 t^r$. Similarly, the constant term of $(1+z)^{a+1}(1+t/z)^a$ is $\sum_r \binom{a+1}{r} \binom{a}{r} t^r$. These are the even and odd cases of (1.1). The last equality follows from (3.3). $\square$

3.2 The Hankel matrix as a Gram matrix

Define a linear map $\Phi : K[x] \rightarrow K[z, z^{-1}]$ on monomials by

$$\Phi(x^{2a}) = A^a, \quad \Phi(x^{2a+1}) = (1+z)A^a. \quad (3.5)$$

This map is not an algebra homomorphism; no multiplicativity of Φ is used. For fixed $k \geq 0$, define the bilinear form

$$B_k(f, g) = \text{CT}_z A^k f(z)g(t/z) \quad \text{on } K[z, z^{-1}]. \quad (3.6)$$

It is symmetric because applying the involution inside the constant term interchanges f and g .

Put $v_i = \Phi(x^i)$. Then

$$B_k(v_i, v_j) = b_{2k+i+j}(t). \quad (3.7)$$

For completeness, if $i = 2a, j = 2b$, the product inside the constant term is A^{k+a+b} . If $i = 2a+1, j = 2b+1$, it is $(1+z)(1+t/z)A^{k+a+b} = A^{k+a+b+1}$. The two mixed-parity cases give $(1+z)A^{k+a+b}$ and $(1+t/z)A^{k+a+b}$, respectively. Lemma 3.2 proves (3.7) in all four cases.

Consequently, $D_{2k}(n, t)$ is the Gram determinant of $v_0, \dots, v_{n-1}$ for the form B_k .

3.3 A triangular Laurent basis change

Consider the ordered monomials

$$w_0 = 1, w_1 = z, w_2 = z^{-1}, w_3 = z^2, w_4 = z^{-2}, \dots; \quad w_{2a} = z^{-a}, w_{2a+1} = z^{a+1}. \quad (3.8)$$

Each v_i belongs to the span of $w_0, \dots, w_i$. The coefficient of the newly appearing monomial is

$$[w_{2a}]v_{2a} = t^a, \quad [w_{2a+1}]v_{2a+1} = 1. \quad (3.9)$$

For the even case, A^a has lowest power z^{-a} , with coefficient t^a , and highest power z^a . For the odd case, $(1+z)A^a$ has highest power z^{a+1} , with coefficient one, and lowest power z^{-a} . This proves both triangularity and (3.9).

Assume $n \geq 1$, and set $q = \lfloor (n-1)/2 \rfloor$. The determinant of the transition matrix from the first n monomials w_i to the first n vectors v_i is therefore

$$\prod_{a=0}^q t^a = t^{q(q+1)/2}.$$

A bilinear Gram determinant changes by the square of this determinant:

$$D_{2k}(n, t) = t^{q(q+1)} \det(B_k(w_i, w_j))_{0 \leq i, j < n}. \quad (3.10)$$

Reorder the monomials increasingly by exponent in both rows and columns. The two permutation signs multiply to one. The resulting exponent set is

$$\begin{cases} \{-m, -m+1, \dots, m\}, & n = 2m+1, \\ \{-(m-1), -(m-2), \dots, m\}, & n = 2m. \end{cases} \quad (3.11)$$

For any two such exponents a, b ,

$$B_k(z^a, z^b) = t^b \text{CT}_z A^k z^{a-b}. \quad (3.12)$$

Factor t^b from the column indexed by b . The sum of the exponents in (3.11) is zero for $n = 2m+1$ and m for $n = 2m$. Combining with (3.10), the total power extracted is

$$\begin{cases} m(m+1), & n = 2m+1, \\ (m-1)m + m = m^2, & n = 2m, \end{cases} = \lfloor n^2/4 \rfloor. \quad (3.13)$$

Since the exponent set is consecutive, the remaining determinant is

$$D_{2k}(n, t) = t^{\lfloor n^2/4 \rfloor} \det(\text{CT}_z A^k z^{i-j})_{0 \leq i, j < n}. \quad (3.14)$$

This identity has been obtained in $\mathbb{Q}(t)$, where all column factors are legitimate, including those with negative exponent.

3.4 Identification of the Toeplitz determinant

We have

$$z^k A(z)^k = (1+z)^k (z+t)^k = \prod_{x \in X_k} (z+x). \quad (3.15)$$

The coefficient of z^{2k-r} in this product is $e_r(X_k)$. Hence

$$\text{CT}_z A^k z^{i-j} = e_{k+i-j}(X_k). \quad (3.16)$$

The conjugate of the rectangle (n^k) is (k^n) . After transposing the determinant, dual Jacobi–Trudi (2.3) gives

$$\det(e_{k+i-j}(X_k))_{0 \leq i, j < n} = s_{(n^k)}(X_k). \quad (3.17)$$

Equations (3.14) and (3.17) prove Theorem 3.1 for $n \geq 1$. For $n = 0$, both determinants and the empty Schur polynomial are one. For $k = 0$, the matrix in (3.17) is the identity matrix, so (1.3) follows as well. Both sides of (3.1) belong to $\mathbb{Z}[t]$; equality in $\mathbb{Q}(t)$ is therefore the claimed polynomial identity. In particular it remains valid at $t = 0$, without dividing a specialized zero by zero. This completes the proof.

4 An explicit positive partition formula

For a partition λ with at most k parts, pad it with zeros and define

$$d_k(\lambda) = \prod_{1 \leq a < b \leq k} \frac{\lambda_a - \lambda_b + b - a}{b - a}. \quad (4.1)$$

This is $s_\lambda(1^k)$, the number of semistandard tableaux of shape λ on k letters, and is a positive integer. The product formula also follows directly from the alternant by letting all variables tend to one: normalized derivatives give a Vandermonde in the exponents $\lambda_{k+1-a} + a - 1$, divided by the Vandermonde in $a - 1$.

Theorem 4.1 (Positive sum of squared tableau counts). *For $k, n \geq 0$,*

$$R_{k,n}(t) = \sum_{\lambda \subseteq (n^k)} d_k(\lambda)^2 t^{|\lambda|}. \quad (4.2)$$

For $k = 0$, the only summand is the empty partition, of dimension one.

Proof. Take a tableau of the rectangle (n^k) on the alphabet $\{1, \dots, 2k\}$. The boxes whose entries are at most k form a Young diagram $\lambda \subseteq (n^k)$. The entries in those boxes have $d_k(\lambda)$ possibilities. The other boxes form the skew shape $(n^k)/\lambda$, and each contributes a factor t .

Define the rectangular complement

$$\lambda^c = (n - \lambda_k, n - \lambda_{k-1}, \dots, n - \lambda_1). \quad (4.3)$$

Rotate the skew tableau through 180 degrees, first subtract k from its entries, and replace each resulting letter a by $k + 1 - a$. Rotation reverses both row and column directions; reversal of the alphabet restores weak increase along rows and strict increase down columns. This is a bijection with tableaux of the ordinary shape λ^c on k letters. Therefore splitting the alphabet gives

$$s_{(n^k)}(1^k, t^k) = \sum_{\lambda \subseteq (n^k)} d_k(\lambda) d_k(\lambda^c) t^{kn - |\lambda|}.$$

The product (4.1), with pairs of indices reversed, shows $d_k(\lambda^c) = d_k(\lambda)$. Replacing λ by λ^c in the sum and using Theorem 3.1 proves (4.2). $\square$

Corollary 4.2 (Polynomiality, full support, and symmetry). *For $k \geq 1, n \geq 0$, every coefficient of $R_{k,n}$ from degree zero through degree kn is a positive integer. Its degree is kn , its constant and leading coefficients are one, and*

$$R_{k,n}(t) = t^{kn} R_{k,n}(t^{-1}). \quad (4.4)$$

Proof. For every $0 \leq r \leq kn$, there exists a partition of size r inside the rectangle: fill as many whole rows of length n as possible and then at most one partial row. The case $n = 0$ is immediate. Each corresponding dimension in (4.2) is positive. Only the empty partition contributes at degree zero, and only the full rectangle contributes at degree kn ; both have dimension one on k letters. Finally $\lambda \mapsto \lambda^c$ preserves the squared dimension and changes the size from r to $kn - r$. $\square$

This argument explains the nonnegativity without extracting coefficients from an alternating determinant. It does *not* by itself prove unimodality: a positive, symmetric sum of monomials need not be unimodal. The next section supplies the missing mechanism.

5 Unimodality by a balanced SL_2 restriction

Lemma 5.1 (Weight strings). *Suppose a finite-dimensional complex algebraic representation of SL_2 has torus character*

$$u^{-N} P(u^2), \quad P(t) = \sum_{r=0}^N a_r t^r.$$

Then there exist nonnegative integers $\mu_0, \dots, \mu_{\lfloor N/2 \rfloor}$ such that

$$P(t) = \sum_{a=0}^{\lfloor N/2 \rfloor} \mu_a t^a (1 + t + \dots + t^{N-2a}). \quad (5.1)$$

In particular the coefficient sequence, including any zero end coefficients, is symmetric and unimodal.

Proof. For clarity, recall why finite-dimensional SL_2 modules have the required strings. Restriction to the compact group SU_2 admits an invariant positive-definite Hermitian form, obtained by averaging any such form. Orthogonal complements of invariant subspaces are invariant; complexifying the Lie algebra gives the same complete reducibility for $\mathfrak{sl}_2$, and hence for the connected group SL_2 .

With generators E, F, H satisfying $[H, E] = 2E$, $[H, F] = -2F$, and $[E, F] = H$, choose a highest-weight vector v in an irreducible summand, so that $Ev = 0$ and $Hv = mv$. The commutator relations imply, by induction,

$$EF^r v = r(m - r + 1)F^{r-1}v.$$

If $F^q v \neq 0$ and $F^{q+1}v = 0$, this formula gives $m = q$. The span of $v, Fv, \dots, F^m v$ is invariant, so irreducibility makes it the whole summand. Its weights are $m, m-2, \dots, -m$, each once. Thus the summand has character $u^{-m} + u^{-m+2} + \dots + u^m$, as does $\mathrm{Sym}^m \mathbb{C}^2$.

In the character assumed in the lemma, all weights lie between $-N$ and N , and all have parity N . Since a direct sum has nonnegative weight multiplicities, every occurring m has the form $N - 2a$, with $0 \leq a \leq \lfloor N/2 \rfloor$. Multiplying the corresponding character by u^N and substituting $t = u^2$ gives the summand in (5.1). All strings have the same center, $N/2$. Their sum is consequently symmetric and weakly increases towards that center. $\square$

Theorem 5.2 (Unimodality). *For all $k \geq 1, n \geq 0$, the polynomial $R_{k,n}(t)$ is unimodal. More precisely, with $N = kn$, it admits (5.1) with $P = R_{k,n}$, and*

$$\mu_a = [t^a]R_{k,n}(t) - [t^{a-1}]R_{k,n}(t) \geq 0 \quad (0 \leq a \leq \lfloor N/2 \rfloor), \quad (5.2)$$

where the coefficient of t^{-1} is zero.

Proof. Identify $\mathbb{C}^{2k}$ with $\mathbb{C}^2 \otimes \mathbb{C}^k$. Let SL_2 act on the first factor and trivially on the second, then restrict the Schur module $S_{(n^k)}(\mathbb{C}^{2k})$ to this subgroup. The torus element $\mathrm{diag}(u^{-1}, u)$ has k eigenvalues u^{-1} and k eigenvalues u on the underlying vector space. Homogeneity of the Schur polynomial and Theorem 3.1 give the restricted character

$$s_{(n^k)}((u^{-1})^k, u^k) = u^{-kn} s_{(n^k)}(1^k, (u^2)^k) = u^{-kn} R_{k,n}(u^2).$$

Apply Lemma 5.1. Up to the center, the coefficient of t^a in (5.1) is $\mu_0 + \dots + \mu_a$, proving (5.2). $\square$

Remark 5.3. The restriction argument applies to $s_\lambda(1^k, t^k)$ for any partition λ with at most $2k$ parts, if the coefficient list is extended through degree $|\lambda|$ by zeros where necessary. The rectangle is needed for the particular Hankel identity and for the nonzero endpoints, not for the general weight-string mechanism. We do not claim this classical representation-theoretic mechanism as new.

6 Explicit numerator polynomials and reciprocity

This section proves the remaining assertions (1.5)–(1.7). All signs and normalizing factorials are included, because the repeated-variable limit is otherwise an easy source of indexing errors.

Fix $k \geq 1$ and $n \geq 0$. Index positions by $I = \{0, 1, \dots, 2k-1\}$, and put

$$\epsilon_a = \begin{cases} a, & 0 \leq a < k, \\ n + a, & k \leq a < 2k, \end{cases} \quad F_k = \prod_{r=0}^{k-1} r!. \quad (6.1)$$

These $2k$ exponents are strictly increasing, including when $n = 0$. For a subset $S \subseteq I$, let ϵ_S denote the list in increasing index order, and write

$$\Delta(\epsilon_S) = \prod_{\substack{a, b \in S \\ a < b}} (\epsilon_b - \epsilon_a).$$

For a k -element subset S , set

$$j(S) = |S \cap \{k, k+1, \dots, 2k-1\}|, \quad (6.2)$$

$$\delta(S) = \sum_{a \in S} a - \binom{k}{2} - j(S)^2, \quad (6.3)$$

$$W_S(n) = \frac{\Delta(\epsilon_S)\Delta(\epsilon_{S^c})}{F_k^2}. \quad (6.4)$$

Finally define

$$C_{k,j}(n, t) = \sum_{\substack{S \subseteq I, |S|=k \\ j(S)=j}} (-1)^{\delta(S)} W_S(n) t^{\delta(S)} \quad (0 \leq j \leq k). \quad (6.5)$$

This is an explicit finite sum with $\binom{2k}{k}$ terms in total.

Lemma 6.1 (Bounds and integrality). *If $j(S) = j$, then*

$$0 \leq \delta(S) \leq 2j(k-j). \quad (6.6)$$

Each endpoint is achieved by a unique such S . Moreover, $W_S(n)$ is a positive integer for every integer $n \geq 0$. As a polynomial in n , it belongs to $\mathbb{Q}[n]$ and has exact degree $2j(k-j)$.

Proof. The smallest sum of indices is obtained by taking the smallest $k-j$ indices from the lower half of I and the smallest j from the upper half. Their sum is $\binom{k}{2} + j^2$. The largest sum, obtained by taking the largest indices in the same two halves, is larger by $2j(k-j)$. These extremizers are unique, proving the bounds.

For increasing nonnegative integers $q_1 < \dots < q_k$,

$$\det \left(\binom{q_b}{r} \right)_{\substack{0 \leq r < k \\ 1 \leq b \leq k}} = \frac{\prod_{a < b} (q_b - q_a)}{F_k}. \quad (6.7)$$

Indeed, the polynomial $\binom{x}{r}$ has degree r and leading coefficient $1/r!$. Changing from these polynomials to powers of x in the determinant proves (6.7). Its left side is an integer, and its right side is strictly positive. Applying it to ϵ_S and ϵ_{S^c} proves the integrality and positivity of $W_S(n)$.

Only differences between one lower-half exponent and one upper-half exponent depend on n . There are $j(k-j)$ such differences in each of the two Vandermonde products. Each has the form $n + c$, with $c > 0$. The claimed degree is therefore exact and its leading coefficient is positive. $\square$

Theorem 6.2 (The conjectured numerator, explicitly). *For all $k \geq 1, n \geq 0$,*

$$(1-t)^{k^2} R_{k,n}(t) = \sum_{j=0}^k (-1)^j t^{j(n+j)} C_{k,j}(n, t). \quad (6.8)$$

For each such n , the polynomials $C_{k,j}(n, t)$ have integer coefficients and exact degree $2j(k-j)$ in t . They satisfy

$$C_{k,k-j}(n, t) = t^{2j(k-j)} C_{k,j}(n, t^{-1}), \quad C_{k,0} = C_{k,k} = 1. \quad (6.9)$$

As a bivariate expression $C_{k,j} \in \mathbb{Q}[n, t]$, its degree in n is also exactly $2j(k-j)$. Thus the requested numerator polynomials are $c_{2k,j} = C_{k,j}$.

6.1 Confluence of the alternant

Use the alternant (2.4) for the partition (n^k) padded to length $2k$. Its increasing exponents are precisely $\epsilon_0, \dots, \epsilon_{2k-1}$. Let the first k variables tend to one and the last k tend to t . Divide numerator and denominator by the within-group Vandermonde products before taking the limit. The normalized derivative of order r of x^{ϵ_a} at $x = y$ is $\binom{\epsilon_a}{r} y^{\epsilon_a - r}$. The remaining denominator is the product of the k^2 cross-group differences, namely $(t - 1)^{k^2}$. Therefore

$$R_{k,n}(t) = \frac{\det M}{(t - 1)^{k^2}}, \quad (6.10)$$

where M is the $2k$ -by- $2k$ matrix with columns indexed by $a \in I$:

$$M_{r,a} = \binom{\epsilon_a}{r}, \quad 0 \leq r < k, \quad (6.11)$$

$$M_{k+r,a} = \binom{\epsilon_a}{r} t^{\epsilon_a - r}, \quad 0 \leq r < k. \quad (6.12)$$

An entry with $r > \epsilon_a$ is zero. This computation is performed in $\mathbb{Q}(t)$, so there is no need to assign a meaning to an intermediate negative power at $t = 0$.

The normalized-derivative rule used here follows by Taylor expansion: when k row variables coalesce at y , the lowest alternating term in their differences is the Vandermonde times the determinant of derivative rows of orders $0, \dots, k - 1$, each divided by its factorial. Thus the calculation does not assume any analytic convergence of infinite series.

6.2 Laplace expansion and its sign

Expand $\det M$ along the bottom k rows, selecting columns S . The top minor is $\Delta(\epsilon_{S^c})/F_k$. Factoring powers of t from the columns and rows of the bottom minor gives

$$\det M_{\text{bottom},S} = t^{\sum_{a \in S} \epsilon_a - \binom{k}{2}} \frac{\Delta(\epsilon_S)}{F_k}. \quad (6.13)$$

By the definitions of j and δ , the exponent is

$$\sum_{a \in S} \epsilon_a - \binom{k}{2} = jn + j^2 + \delta(S). \quad (6.14)$$

Using zero-based row and column indices, the Laplace sign is $(-1)^{k(3k-1)/2 + \sum_{a \in S} a}$. Changing the denominator in (6.10) from $(t - 1)^{k^2}$ to $(1 - t)^{k^2}$ adds the factor $(-1)^{k^2}$. The resulting sign simplifies as follows:

$$\begin{aligned} (-1)^{k^2 + k(3k-1)/2 + \sum_{a \in S} a} &= (-1)^{\binom{k}{2} + \sum_{a \in S} a} \\ &= (-1)^{j^2 + \delta(S)} = (-1)^{j + \delta(S)}. \end{aligned} \quad (6.15)$$

Grouping subsets according to $j(S)$ proves (6.8).

6.3 Exact degrees and complementary subsets

Lemma 6.1 shows that $C_{k,j}$ is an integer polynomial in t , with degree at most $d = 2j(k - j)$. The unique subset at $\delta = d$ contributes the nonzero coefficient $(-1)^d W_S(n) = W_S(n) > 0$. Hence the degree is exactly d , for every nonnegative integer n . The unique subset at $\delta = 0$ likewise shows that the coefficient of n^d in $C_{k,j}$, as a polynomial in t , has nonzero constant term. This proves the exact degree in n .

Complementing the selected subset gives

$$j(S^c) = k - j(S), \quad \delta(S^c) = 2j(k - j) - \delta(S), \quad W_{S^c}(n) = W_S(n). \quad (6.16)$$

Since $2j(k - j)$ is even, the signs of complementary terms agree. Equation (6.9) follows immediately. If $j = 0$ or $j = k$, there is only one subset, both Vandermonde products are those of consecutive integers, and their quotient by F_k^2 is one. This completes the proof of Theorem 6.2.

Remark 6.3 (Integer-valued is not the same as integral coefficients in n). The formula gives $C_{k,j}(n, t) \in \mathbb{Z}[t]$ at every integer $n \geq 0$, but we only assert $C_{k,j} \in \mathbb{Q}[n, t]$ before specializing n . The Vandermonde quotients are integer-valued polynomials. No claim that their ordinary monomial coefficients in n are all integers is needed. Also, the $C_{k,j}$ generally have alternating signs in t ; positivity in the conjecture concerns $R_{k,n}$, not these numerator blocks.

Corollary 6.4 (Resolution of the selected conjecture). *Conjecture 8 of Cigler's paper [1] holds for every $k \geq 1, n \geq 0$. Its polynomiality, positive full support, exact degree, palindromicity, unimodality, numerator representation, numerator degrees, and numerator reciprocity are established respectively by Theorem 3.1, Corollary 4.2, Theorem 5.2, and Theorem 6.2.*

7 Coefficient edges and special values

Proposition 7.1 (Stable initial coefficients). *For $k \geq 1$ and $0 \leq r \leq n$,*

$$[t^r]R_{k,n}(t) = \binom{k^2 + r - 1}{r}. \quad (7.1)$$

The next coefficient, with coefficients outside the degree interpreted as zero, is

$$[t^{n+1}]R_{k,n}(t) = \binom{k^2 + n}{n+1} - \binom{n+k}{k-1}^2. \quad (7.2)$$

Proof. The $j = 0$ term in (6.8) is one; every other term starts at degree at least $n + 1$. Thus

$$R_{k,n}(t) \equiv (1 - t)^{-k^2} \pmod{t^{n+1}},$$

which proves (7.1) by the binomial series. At degree $n + 1$, only the constant coefficient of $C_{k,1}$ supplies a correction. Its unique subset has lower indices $0, \dots, k - 2$ and upper index k . Evaluation of (6.4) gives

$$C_{k,1}(n, 0) = \left(\frac{(n+2)(n+3) \cdots (n+k)}{(k-1)!} \right)^2 = \binom{n+k}{k-1}^2.$$

Empty products cover $k = 1$. Substitution in (6.8) proves (7.2). $\square$

Proposition 7.2 (Dimension product and monotonicity in the order). *For $k, n \geq 0$,*

$$R_{k,n}(1) = \prod_{i=1}^k \prod_{j=1}^k \frac{n+i+j-1}{i+j-1}. \quad (7.3)$$

Also, $R_{k,n+1}(t) - R_{k,n}(t)$ has nonnegative coefficients, and $R_{k,n}(0) = 1$.

Proof. At $t = 1$, Theorem 3.1 gives $s_{(n^k)}(1^{2k})$. In the dimension formula (4.1), now with $2k$ letters, factors from pairs within the same half cancel. The remaining factors are

$$\prod_{a=1}^k \prod_{b=k+1}^{2k} \frac{n+b-a}{b-a}.$$

Set $i = k + 1 - a$ and $j = b - k$ to obtain (7.3). The rectangles (n^k) form an increasing family, and the summand $d_k(\lambda)^2 t^{|\lambda|}$ in (4.2) is independent of n . This proves coefficientwise monotonicity. The last assertion was already proved in Corollary 4.2. $\square$

The apparent poles at $t = 1$ in (6.8) are therefore removable. For the original determinant, $D_{2k}(n, 1) = R_{k,n}(1)$. At $t = 0$, the original determinant is zero for $n \geq 2$, while its normalized polynomial has value one. These are different statements and should not be conflated.

The parameter t records how many tableau entries are in the upper half of the alphabet. This is a repeated-variable specialization, not the principal specialization $1, t, t^2, \dots$; standard volume-generating formulas for plane partitions cannot simply be substituted for it.

8 A generic recurrence and three asymptotic regimes

The explicit numerator has information beyond coefficient shape. The following results concern fixed k and varying matrix order n .

Theorem 8.1 (Exact generic recurrence). *Fix $k \geq 1$, and regard t as an indeterminate. Let E be the forward shift in n . Over $\mathbb{Q}(t)$, the minimal monic constant-coefficient recurrence operator for $R_{k,n}(t)$ is*

$$\prod_{j=0}^k (E - t^j)^{2j(k-j)+1}. \quad (8.1)$$

Equivalently, the generating function in the order n has reduced denominator

$$\prod_{j=0}^k (1 - t^j z)^{2j(k-j)+1}. \quad (8.2)$$

Its degree in z , and the recurrence order, is

$$L_k = \sum_{j=0}^k (2j(k-j) + 1) = \frac{(k+1)(k^2 - k + 3)}{3}. \quad (8.3)$$

Proof. Rewrite (6.8) in the form

$$R_{k,n}(t) = \sum_{j=0}^k t^{jn} Q_j(n, t), \quad Q_j(n, t) = \frac{(-1)^j t^{j^2} C_{k,j}(n, t)}{(1-t)^{k^2}}.$$

By Theorem 6.2, Q_j has exact degree $d_j = 2j(k-j)$ in n . Finite differences give $(E - t^j)^{d_j+1}(t^{jn} Q_j(n, t)) = 0$; multiplying all these commuting operators proves the recurrence.

For minimality, recall that the generating function of a polynomial in n of exact degree d has a pole of exact order $d+1$ at one. This follows by applying $(w d/dw)^d$ to $(1-w)^{-1}$: the coefficient of its highest-order pole is $d!$, times the leading coefficient of the polynomial. The j -th summand above consequently has a pole of exact order $d_j + 1$ at $z = t^{-j}$. Over $\mathbb{Q}(t)$ these locations are distinct, so no other summand cancels that pole. This proves the exact denominator and hence minimality. Summing the exponents gives (8.3). $\square$

The denominator is the one appearing in the nearby Conjecture 10 of [1]. That conjecture also specifies a numerator degree and a separate reciprocity identity, which are not asserted here. The present article's complete-resolution claim concerns Conjecture 8 only. For specialized values of t , poles can coincide or disappear, so Theorem 8.1 must not be read as an unchanged minimal-order claim after every specialization. At $t = 0$, the sequence is constant; at $t = 1$, it is a polynomial in n of degree k^2 .

Theorem 8.2 (Fixed-parameter asymptotics). *Fix $k \geq 1$. For every fixed real t with $0 < t < 1$,*

$$R_{k,n}(t) \longrightarrow (1-t)^{-k^2} \quad (n \rightarrow \infty), \quad (8.4)$$

and the deficit has the sharp leading term

$$(1-t)^{-k^2} - R_{k,n}(t) \sim \frac{n^{2k-2}t^{n+1}}{((k-1)!)^2(1-t)^{k^2-2k+2}}. \quad (8.5)$$

At $t = 1$,

$$R_{k,n}(1) \sim \frac{n^{k^2}}{\prod_{i=1}^k \prod_{j=1}^k (i+j-1)}. \quad (8.6)$$

For fixed $t > 1$,

$$R_{k,n}(t) \sim t^{kn}(1-t^{-1})^{-k^2}. \quad (8.7)$$

Proof. For $0 < t < 1$, every term with $j \geq 1$ in the numerator formula vanishes as $n \rightarrow \infty$, because it is a fixed-degree polynomial in n times t^{jn} . This proves (8.4).

To identify the first error term, compute the leading coefficient in n of $C_{k,1}$. Put $m = k - 1$. A selected subset with one upper-half index consists of all lower-half indices except some a , and the single upper index $k + b$, where $0 \leq a, b \leq m$. Its δ is $m - a + b$. Deleting one element from a consecutive Vandermonde gives

$$[n^{2m}]W_S(n) = \frac{\binom{m}{a}\binom{m}{b}}{(m!)^2}.$$

Consequently,

$$\begin{aligned} [n^{2k-2}]C_{k,1}(n, t) &= \frac{1}{(m!)^2} \sum_{a,b=0}^m \binom{m}{a} \binom{m}{b} (-t)^{m-a+b} \\ &= \frac{(1-t)^{2m}}{(m!)^2}. \end{aligned}$$

The $j = 1$ term in (6.8) is $-t^{n+1}C_{k,1}$. All terms with $j \geq 2$ are exponentially smaller than this term for fixed $t \in (0, 1)$. Division by $(1-t)^{k^2}$ proves (8.5). For $k = 1$, the error formula is exact: it is $t^{n+1}/(1-t)$.

Equation (8.6) follows from the product in (7.3). Finally, palindromicity reduces the case $t > 1$ to (8.4) with parameter t^{-1} . $\square$

These are fixed- t assertions, not estimates uniform as $t \rightarrow 1$. A transition regime with t depending on n would require a separate analysis.

9 Worked examples and explicit checks on conventions

For $k = 1$, the rectangle has one row. Either the positive formula or Jacobi–Trudi gives

$$R_{1,n}(t) = 1 + t + \cdots + t^n = \frac{1-t^{n+1}}{1-t}. \quad (9.1)$$

Here $C_{1,0} = C_{1,1} = 1$, and the numerator formula is immediate. This family also shows why “unimodal” cannot be strengthened to “strictly unimodal” without exceptions. For $n = 2$, the roots of $1 + t + t^2$ are nonreal; real-rootedness is not a premise of our proof.

For $k = 2$, evaluation of (6.5) gives

$$\begin{aligned} C_{2,0} &= C_{2,2} = 1, \\ C_{2,1}(n, t) &= (n+2)^2 - 2(n^2 + 4n + 3)t + (n+2)^2 t^2. \end{aligned} \quad (9.2)$$

Thus

$$R_{2,n}(t) = \frac{1 - t^{n+1}((n+2)^2 - 2(n^2 + 4n + 3)t + (n+2)^2t^2) + t^{2n+4}}{(1-t)^4}. \quad (9.3)$$

Cigler had already obtained this small-shift formula in his equation (53); it is included as a normalization check, not claimed as a new special case. The contribution here is the all- k mechanism and its consequences.

Some small normalized polynomials are:

k	n	$R_{k,n}(t)$
1	2	$1 + t + t^2$
2	1	$1 + 4t + t^2$
2	2	$1 + 4t + 10t^2 + 4t^3 + t^4$
2	3	$1 + 4t + 10t^2 + 20t^3 + 10t^4 + 4t^5 + t^6$
3	1	$1 + 9t + 9t^2 + t^3$
3	2	$1 + 9t + 45t^2 + 65t^3 + 45t^4 + 9t^5 + t^6$

For $k = 3, n = 3$, the coefficient list is

$$(1, 9, 45, 165, 270, 270, 165, 45, 9, 1).$$

Its weight-string multiplicities up to the center are $(1, 8, 36, 120, 105)$, the consecutive differences prescribed by (5.2).

As a direct two-by-two check on the Hankel normalization,

$$\begin{aligned} D_2(2, t) &= (1+t)(1+4t+t^2) - (1+2t)^2 \\ &= t + t^2 + t^3 = t^{\lfloor 2^2/4 \rfloor} R_{1,2}(t). \end{aligned}$$

The factor t is essential; omitting it would make both the Schur identity and the stated coefficient degree wrong.

10 Verification, dependencies, and limits of the claim

10.1 What the executable verifier does

The accompanying file `code/verify.py` uses only Python's standard library and arbitrary-precision integers. Polynomials are stored as coefficient tuples in ascending degree. It implements addition, multiplication, and exact polynomial division, with explicit checks for zero remainders and integral quotients. The determinant engine is the fraction-free Bareiss algorithm with row pivoting; every intermediate exact division is checked.

For all $0 \leq k \leq 6$ and $0 \leq n \leq 10$, it compares the following formulas:

- (1) the original shifted Hankel determinant divided exactly by $t^{\lfloor n^2/4 \rfloor}$;
- (2) the elementary-symmetric Toeplitz determinant $\det(e_{k+i-j}(X_k))$;
- (3) the complete-symmetric Jacobi–Trudi determinant $\det(h_{n-i+j}(X_k))_{0 \leq i, j < k}$;
- (4) the positive rectangular partition sum (4.2);
- (5) for $k \geq 1$, the subset numerator (6.5) reconstructed through exact division by $(1-t)^{k^2}$.

The three determinant formulations share the Bareiss implementation; they are independent formulas, not three independent software systems. As a separate audit, every original Hankel determinant of order at most four in the tested grid is also calculated by the signed-permutation Leibniz formula, which does not use Bareiss elimination.

The verifier also checks degree, positivity of every coefficient, palindromicity, weak unimodality, the dimension product at $t = 1$, the stable leading coefficients, coefficientwise monotonicity in n , and the exact degrees and reciprocity of the numerator blocks. The saved run reported:

Check category	Count
Parameter pairs (k, n)	77
Determinant-to-partition comparisons	231
Numerator reconstructions	66
Independent signed-permutation audits	35
Disagreements	0

The run used Python 3.13.5. Exact outputs are saved in `results/verification.json`, `results/polynomials.json`, and `results/numerator_blocks.json`. To reproduce the default grid, run from the package directory:

```
python3 code/verify.py
```

Larger bounds can be selected with `-max-k` and `-max-n`. Enumeration of partitions is finite but grows with the rectangle; the default run is deliberately a check rather than a large-scale search. No floating-point computation, numerical root test, or guessed recurrence is used to establish an identity.

The supplementary script `code/verify_consequences.py` adds 66 exact checks of (7.2), six symbolic checks of the leading coefficient controlling (8.5), and nine recurrence windows for $k = 1, 2, 3$, computed from original Hankel determinants up to order 14. All passed. The leading coefficients are checked by exact finite differences in n , not by numerical fitting. The output is `results/consequences.json`; reproduce it with

```
python3 code/verify_consequences.py
```

10.2 Proof audit

The roles of the main ingredients are distinct. The moment identity and basis change prove the full Hankel-to-Toeplitz reduction, including the normalization. Dual Jacobi–Trudi identifies the resulting determinant. The tableau decomposition proves full support and palindromicity without representation theory. The Schur-module character theorem and the proved weight-string lemma give unimodality. The confluent alternant and its Laplace expansion prove the numerator assertions, including their signs and exact degrees. The generic recurrence and asymptotics are consequences of those explicit formulas.

In particular, no induction step divides by a shifted Hankel determinant that might vanish at a special parameter value. Intermediate calculations in $\mathbb{Q}(t)$ are converted to polynomial identities before specialization. The proof treats both parities of n , the empty determinant, $k = 0$, $n = 0$, and the apparent singularities at $t = 0$ and $t = 1$.

The imported mathematical facts are the tableau/Schur identities (2.2)–(2.4) and the classical Schur-module character theorem. These are standard theorems with references, not unproved conjectures or results borrowed from the supplied manifest. The finite checks are supplemental and cannot replace the all-parameter proof.

10.3 What is not asserted

This article does not settle the odd-shift Conjecture 9, the remaining numerator assertions of Conjecture 10, or Conjectures 13–15 concerning the separate family catalogued in the manifest. It makes no general real-rootedness or strict-unimodality assertion. It does not claim that the small examples already printed by Cigler are new.

The proof has been derived and checked within this research session, but there has been no external referee report and no proof-assistant formalization. The source version explicitly formulates the selected problem as a conjecture, and the search did not locate a prior resolution; neither fact rules out an unpublished proof, a differently indexed result, or a resolution outside the

sources searched. The mathematical statement and its argument should be assessed independently of any priority claim.

11 Conclusion

The even-shift middle-binomial Hankel determinant conceals a particularly simple Toeplitz determinant. The parity-sensitive map $x^{2a} \mapsto A^a$, $x^{2a+1} \mapsto (1+z)A^a$, together with the involution $z \mapsto t/z$, exposes it. Once the precise power $t^{\lfloor n^2/4 \rfloor}$ is accounted for, the normalized answer is a rectangular Schur polynomial at two equally repeated values.

That one identification separates the conjecture into three transparent mechanisms: tableaux for positive symmetric coefficients, SL_2 strings for unimodality, and complementary minors of a confluent alternant for the rational numerator and its reciprocity. Each part works for all k and n ; none is an extrapolation from the finite test grid. This supplies a complete proof draft of the selected, nonduplicate Conjecture 8, together with explicit formulas and reproducible checks.

References

- [1] Johann Cigler, *Hankel determinants of middle binomial coefficients and conjectures for some polynomial extensions and modifications*, arXiv:2111.14492v3, 30 December 2021. Section 4, Conjecture 8, printed pp. 16–17; equations (55)–(57). <https://arxiv.org/abs/2111.14492v3>. Accessed 20 September 2026.
- [2] William Fulton, *Young Tableaux: With Applications to Representation Theory and Geometry*, London Mathematical Society Student Texts 35, Cambridge University Press. Parts I–II; in particular the Schur-module character statement in the introduction to Part II, pp. 79–82. Online chapter DOI: <https://doi.org/10.1017/CB09780511626241.010>. Publisher’s character-theorem summary accessed 20 September 2026.
- [3] I. G. Macdonald, *Symmetric Functions and Hall Polynomials*, second edition, Oxford University Press, 1995. Chapter I, especially the Schur-function identities and specializations.
- [4] Vladimir Reshetnikov, *A manifest of docs/reports*, 20 September 2026. User-supplied L^AT_EX file `manifest.tex`, catalogue of 71 research packages. The related entry is “Product formulas for ballot-polynomial Hankel determinants”, concerning Cigler’s Conjectures 13–15. Used for topic selection and duplicate avoidance, not as a proof source.